Broad Sound Taxonomy Classification with Textual Metadata and Crowdsourced Annotations

Paul Primus[1], *Pao Lin*[1], *Gerhard Widmer*[1,2]

[1]Institute of Computational Perception, Johannes Kepler University Linz, Austria
[2]LIT Artificial Intelligence Lab, Linz, Austria
first.last@jku.at

***Abstract*—The Broad Sound Taxonomy (BST) provides a compact hierarchy for organizing arbitrary sound recordings into 5 top-level and 23 second-level categories. The DCASE 2026 Challenge Task 1, *Heterogeneous Audio Classification*, evaluates systems for automatically categorizing recordings from the Freesound platform into the correct second-level BST category based on audio and textual metadata. In this work, we investigate a metadata-aware classification approach that uses a large language model to predict BST class probabilities from clip metadata and fuses these predictions with audio–text embeddings. In addition, we study strategies for exploiting a larger crowdsourced and noisily annotated training set. To this end, we generate soft pseudo-labels from models trained on clean data and use them as training targets for the noisy subset. We evaluate our approach on the BSD10k dataset in the context of DCASE 2026 Task 1 using M2D- and CLAP-based classifiers. Our results suggest that using LLM-derived class predictions is complementary to processing metadata only with a pretrained CLAP text encoder. For using a larger dataset with crowdsourced annotations we find that training on soft pseudo-labels is more effective than using the noisy labels, but that our simple pseudo-labeling strategy does not improve over training on the clean data alone; performance appears to saturate even as the pseudo-labeled set grows. Finally, error analysis shows that our model's class confusions are concentrated in classes that also exhibit human annotation ambiguity, suggesting that further progress may require sharper BST class definitions in addition to improved audio–text modeling.**

***Index Terms*—Heterogeneous audio classification, Broad Sound Taxonomy, audio–text classification, noisy-label learning**

1. INTRODUCTION

The Broad Sound Taxonomy (BST) [1] organizes heterogeneous audio into a two-level hierarchy with five top-level categories—music, instrument samples, speech, sound effects, and soundscapes—and 23 more specific second-level classes. Since 2025, BST has also been integrated into Freesound [2], where it supports the organization and retrieval of recordings. Freesound recordings are also accompanied by textual metadata describing sound sources, recording environments, or broader context, which provides complementary information.

In this work, we consider the task of automatically assigning Freesound items to their corresponding second-level BST categories using both the audio signal and its metadata. This setting motivates exploring audio–text models such as CLAP [3–5], which allow representing audio and metadata in common embedding space.

Training audio–text classifiers, however, requires large datasets with reliable annotations, whose curation is costly and time-consuming. Freesound provides a large collection of user-contributed recordings and metadata and, in principle, also enables crowdsourced BST annotation, for example by asking users to select a category when uploading a recording. However, such annotations are often noisy. Effective use of this data therefore requires methods that exploit the additional recordings without directly learning their label errors. Prior work on heterogeneous and web-audio classification has demonstrated both the value and limitations of Freesound-derived annotations [6–9].

In this work, we focus on two challenges of classifying heterogeneous Freesound recordings into the BST: exploiting textual metadata and learning from noisy crowdsourced annotations. Our experiments use the smaller, carefully curated BSD10k dataset [10, 11] and the larger BSD35k-CS dataset with crowdsourced annotations [12].

The paper is organized accordingly. Section 2 investigates metadata-aware BST classification. We use GPT-5.4-mini [13] to derive class probabilities from the metadata and BST definitions, map them to a learned class-embedding representation, and fuse this representation with audio and text embeddings. Section 3 examines how BSD35k-CS can be exploited despite its noisy labels. We replace the original annotations with soft pseudo-labels [14, 15] generated by models trained on BSD10k, pretrain on the resulting pseudo-labeled data, and subsequently fine-tune on clean BSD10k. Section 4 describes the experimental setup, and Section 5 presents the results.

Our experiments show that LLM-derived class predictions consistently improve BST classification and that soft pseudo-labels provide substantially better supervision than the original crowdsourced annotations. We further analyze the effects of pseudo-label weighting and training-set size, as well as the most frequent model errors. The resulting systems were submitted to the DCASE 2026 Challenge, where our submissions [16] ranked first among 41 entries.

2. METADATA-AWARE AUDIO CLASSIFICATION

This section first assesses the predictive value of the metadata and then introduces two complementary metadata representations: a continuous CLAP text embedding and an embedding derived from LLM-predicted class probabilities. Finally, we describe how these representations are fused with audio embeddings for classification.

2.1. Metadata Analysis

Each Freesound recording includes three user-provided metadata fields: a title, a set of tags, and a free-form description. Titles are typically short and may directly identify the main sound source or event, such as an instrument, animal, vehicle, or environment. Tags provide compact category cues, but vary considerably in number and specificity and may be missing or ambiguous. Descriptions are often the most informative field, providing details about sound sources, recording environments, procedures, or intended use. However, they may also contain less relevant material, including licensing statements, recording device information, attribution, links, or HTML markup.

To quantify the informativeness of the metadata for BST class prediction, we train two unimodal classifiers: one using the LAION-CLAP audio encoder and the other using its text encoder. On the held-out test split, the text-only model, which uses an LLM-generated summary of the metadata as input, achieves a class-balanced accuracy of $70.38\% \pm 0.115$, while the audio-only model achieves 70.25%. Thus, metadata alone performs comparably to audio, demonstrating that it provides a strong independent signal for BST classification and motivating the fusion of both modalities.

2.2. Metadata Preprocessing and CLAP Encoding

Concatenating the title, tags, and description yields text of variable length and structure. These long strings may include non-acoustic information and exceed the effective context of text encoders pretrained on short audio captions. We therefore use GPT-5.4-mini [13] to transform the raw metadata into a short, caption-style description of the expected audio content.

For each recording, the LLM receives the title, tags, and description and produces a concise one- or two-sentence summary containing only information relevant to the BST category. The prompt emphasizes clear descriptions of the sound content and removes promotional, licensing, attribution, and technical information. The summaries are generated once and cached. The full prompt and resulting summaries are available in our repository.[1]

The resulting summary is then projected to a single embedding vector using a pre-trained text encoder $f_{\mathrm{t}}^{\mathrm{CLAP}}(s_i)$. Let s_i denote the resulting metadata summary for recording i, then

$$u_i = f_{\mathrm{t}}^{\mathrm{CLAP}}(s_i) \in \mathbb{R}^{D_{\mathrm{t}}} \tag{1}$$

is the text embedding where D_{t} denotes its dimensionality. We choose the LAION-CLAP [4] pretrained RoBERTa text encoder [17] and use the encoder's output before the unit-norm normalization, which is typically applied for contrastive audio–text similarity.

2.3. Metadata-Based Class Prediction with an LLM

We additionally experiment with a representation that is more explicitly aligned with the 23 BST target classes and derived using an LLM conditioned on textual class descriptions. For each clip, GPT-5.4-mini with reasoning effort set to medium is prompted to estimate a probability distribution over plausible BST classes using the unprocessed metadata and the duration of the recording. The prompt contains the complete set of allowed class labels together with class definitions and representative examples. The LLM is instructed to select the classes according to the class definitions and return a set of plausible labels. Its response must be valid JSON containing class labels and probability values. As with caption generation, the LLM predictions are generated offline and cached. The full prompt, including the class definitions and formatting requirements, together with the full output, is provided in the repository.

Although the resulting class prediction could be concatenated directly with the backbone representation, its $C = 23$ dimensions have a fixed class-specific interpretation and a substantially lower dimensionality than the backbone embeddings. We therefore learn a continuous representation of the class prediction. Let

$$q_i \in \mathbb{R}^C \tag{2}$$

denote the probability distribution returned by the LLM over the $C = 23$ target classes, where the probabilities sum to one. This vector constitutes the metadata-derived class prediction. Further let

$$E \in \mathbb{R}^{C \times D_{\mathrm{m}}} \tag{3}$$

be a trainable class-embedding matrix, where row E_c represents an embedding vector corresponding to BST class c of dimensionality D_m. The metadata-prediction embedding is the probability-weighted combination

$$m_i = q_i^\top E = \sum_{c=1}^{C} q_{i,c} E_c, \qquad m_i \in \mathbb{R}^{D_{\mathrm{m}}}. \tag{4}$$

This operation maps the sparse, class-aligned LLM output into a dense representation learned jointly with the final classifier.

[1] https://github.com/OptimusPrimus/dcase2026-task1

2.4. Audio–Metadata Fusion

The audio is processed with a pre-trained audio encoder. Let x_i denote the audio waveform of recording i. For an arbitrary audio encoder f_{a}, the audio embedding is

$$a_i = f_{\mathrm{a}}(x_i) \in \mathbb{R}^{D_{\mathrm{a}}}, \tag{5}$$

where D_{a} depends on the selected backbone. We use two alternatives: the audio encoder of LAION-CLAP [4], which is based on tiny-HTSAT [18], and M2D [19]. As with the CLAP text representation, we use the CLAP audio encoder output before unit-norm normalization.

The final model combines three sources of information for every Freesound item: the audio embedding, a CLAP text embedding computed from the LLM-generated metadata caption, and an embedding of the LLM-derived BST class prediction:

$$z_i = [a_i; u_i; m_i] \in \mathbb{R}^{D_{\mathrm{a}}+D_{\mathrm{t}}+D_{\mathrm{m}}}, \tag{6}$$

where $[\cdot;\cdot;\cdot]$ denotes vector concatenation. The fused representation is processed by a shallow fully connected network

$$\ell_i = g_\theta(z_i), \tag{7}$$

where g_θ denotes the fusion network and $\ell_i \in \mathbb{R}^C$ contains the logits for the $C = 23$ second-level BST classes. A softmax is applied to obtain class probabilities, and the network is trained using cross-entropy loss.

3. SOFT PSEUDO-LABEL GENERATION FOR BSD35K-CS

BSD35k-CS contains roughly three times as many recordings as the cleanly annotated BSD10k dataset, but its crowdsourced labels are substantially noisier. This section investigates whether and how these additional recordings can be exploited using soft pseudo-labels from an ensemble trained on BSD10k. The resulting three-stage procedure consists of clean-data training, pseudo-labeled pretraining on BSD35k-CS, and final fine-tuning on BSD10k.

3.1. Annotation Analysis of BSD35k-CS

BSD35k-CS is the larger of the two Freesound datasets and contains 33,829 recordings, compared with 10,956 recordings in BSD10k. Both datasets use the same 23 second-level BST classes and provide the same audio and textual metadata modalities. Their difference is the expected reliability of their annotations. BSD10k was carefully curated and cleanly annotated, whereas the labels in BSD35k-CS were obtained through crowd sourcing and may therefore contain incorrect class annotations.

To quantify annotation noise, we compare two otherwise identical CLAP-based classifiers trained on either clean BSD10k or BSD35k-CS with its original crowdsourced labels. Both are evaluated on the held-out BSD10k test split. The BSD10k-trained model achieves 76.5% balanced accuracy, whereas the BSD35k-CS-trained model reaches only 55.08%. Despite using more data, training on BSD35k-CS performs substantially worse, suggesting that hard crowdsourced targets encourage the model to learn annotation errors. This motivates using predictions from clean-data models as an alternative supervision signal.

3.2. Clean-Data Training (Stage 1)

In the first stage, we train a set of multimodal classifiers with different audio embedding backbones on the BSD10k training split using the provided ground-truth class labels. Each model processes the same audio–metadata input but differs in its audio feature extractor, allowing the ensemble to capture complementary acoustic information. For an input example x_i,

$$\ell_{i,m} \in \mathbb{R}^C$$

denotes the vector of pre-softmax logits predicted by model m, where C is the number of target classes.

3.3. Pseudo-Labeled Pretraining on BSD35k-CS (Stage 2)

To obtain better prediction targets for BSD35k-CS samples, we average the logits of M selected Stage 1 models ($M = 2$) and apply a softmax transformation.

$$\tilde{\mathbf{y}}_i = \mathrm{softmax}\left(\frac{1}{M}\sum_{m=1}^{M} \ell_{i,m}\right). \tag{8}$$

The resulting vector $\tilde{\mathbf{y}}_i \in [0, 1]^C$ is stored as a soft pseudo-label for example i which we use directly for training. The use of a soft ensemble distribution transfers information about relations among plausible classes that would be absent from a one-hot target [15].

In our setup, each BSD35k-CS training example has two potential supervision signals: its original crowdsourced one-hot label and the soft distribution predicted by the Stage-1 ensemble. We therefore compute two cross-entropy losses: one against the original crowdsourced label and one against the soft pseudo-label distribution predicted by the Stage-1 ensemble. The two objectives are combined as

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_{\text{supervised}} + \lambda\mathcal{L}_{\text{pseudo}}, \tag{9}$$

where $\lambda \in [0, 1]$ determines the relative influence of the teacher ensemble. For Stage 2, λ is treated as a hyperparameter.

3.4. Clean-Data Fine-Tuning (Stage 3)

In the final stage, the resulting Stage-2 model is fine-tuned on BSD10k using only the original ground-truth annotations. The objective is therefore the standard supervised cross-entropy loss. This final fine-tuning step readjusts the decision boundaries toward the trusted label distribution after pretraining on BSD35k-CS. It is intended to preserve representations learned from the larger dataset while correcting biases introduced by residual errors in the pseudo-labels or by differences between the BSD10k and BSD35k-CS data distributions.

4. EXPERIMENTAL SETUP

Following a data-partitioning strategy similar to the DCASE Task 1 baseline, we divide BSD10k into an 80/20 development and test split. The test split is held out for reporting results there, while the development set is split into training and validation subsets using an 80/20 ratio. The validation subset is used for checkpoint selection and hyperparameter tuning (learning rate, number of epochs, and pseudo-label loss weight λ). BSD35k-CS is used only for Stage 2 training. We report macro-averaged hierarchical precision, recall, and F1-score (hP, hR, and hF1) averaged over three random seeds. These metrics assign full credit to exact second-level BST matches and partial credit of $\lambda_{\mathrm{h}} = 0.375$ when the predicted and target classes differ but share the same top-level BST category. Since hF1 is the primary evaluation metric in the DCASE challenge, we focus mainly on this measure. Full metric definitions are provided on the DCASE 2026 Task 1 website [20]. Training scripts and technical details are available in our repository.

5. EXPERIMENTS

This section evaluates the proposed metadata-fusion and pseudo-labeling methods. We first compare the CLAP-like audio–text classifiers (using M2D or HTSAT for audio encoding) with our proposed architecture which additionally fuses the LLM-derived class predictions before final classification. We then examine the three-stage training strategy, including the effect of replacing the crowdsourced annotations with soft pseudo-labels and varying the amount of pseudo-labeled training data. Finally, we analyze the resulting model's classification errors in more detail.

5.1. Main Results

Table 1 summarizes the effect of the metadata-derived LLM class predictions (Stage 1), the pseudo-label training (Stage 2), and the subsequent fine-tuning (Stage 3). In Stage 1, incorporating the LLM predictions consistently improves both backbones. For CLAP, mean hF1 increases from 78.65 to 80.96, a gain of 2.31 percentage points. For M2D, hF1 increases from 78.75 to 80.56, corresponding to a gain of 1.81 points. These results suggest that the explicit BST-class predictions extracted from the metadata using the class definitions and an LLM provides valuable information that is complementary to the dense CLAP text encoder representation.

The Stage-1 ensemble of M2D and CLAP with LLM class predictions obtains an hF1 of 81.63, outperforming the individual Stage-1 models. Training a CLAP model on BSD35k-CS with the soft pseudo-labels from the Stage-1 ensemble yields an hF1 of 81.22. Although this exceeds the corresponding Stage-1 CLAP model's 80.96, it does not surpass the Stage-1 ensemble that generated the pseudo-labels. Thus, pseudo-label training alone does not improve on the clean-data system. Rather, the student approximately recovers the performance of its teacher ensemble in a single model.

Fine-tuning the Stage-2 model on BSD10k in Stage 3 produces the highest mean hF1, 82.04. However, the improvement over the Stage-1 ensemble is only marginal with 0.41 percentage points.

Table 1: Mean performance (%) over three runs. Error bars indicate the sample standard deviation. The audio encoder column specifies the model used to extract audio embeddings; the RoBERTa-based CLAP text encoder is used in all configurations. LLM indicates whether the metadata-derived LLM class predictions are included. The ensemble averages the predictions of the HTSAT- and M2D-based models with LLM input.

Stage	Audio Encoder	LLM	hF1	hP	hR
1	HTSAT	×	78.65 ± 0.71	79.93	77.77
	M2D	×	78.75 ± 0.55	79.56	78.29
	HTSAT	✓	80.96 ± 0.24	82.39	80.11
	M2D	✓	80.56 ± 0.54	81.29	80.18
	HTSAT + M2D	✓	81.63 ± 0.37	—	—
2	HTSAT	✓	81.22 ± 0.53	82.22	80.58
3	HTSAT	✓	$\mathbf{82.04 \pm 0.33}$	**82.64**	**81.74**

5.2. Pre-training with noisy annotations

To assess the influence of the original crowdsourced annotations, we vary the pseudo-label loss weight λ in Stage 2 from 0 to 1 in increments of 0.2. At $\lambda = 0$, training uses only the original BSD35k-CS annotations, whereas $\lambda = 1$ discards them and uses only the soft pseudo-labels produced by the Stage-1 ensemble. The resulting BSD10k validation hF1 scores are shown in Fig. 1.

Performance improves steadily as more weight is assigned to the pseudo-labels, with the best results obtained when training is dominated by or relies entirely on the ensemble predictions. In contrast, greater reliance on the original crowdsourced annotations substantially degrades performance.

These results suggest that pseudo-labels from our clean-data models provide more useful supervision than the original crowdsourced annotations. However, this conclusion may differ when fewer clean training examples are available. The main benefit of pseudo-label training may instead arise from the greater variability introduced by the additional audio recordings and their associated metadata. Given the large number of unlabeled recordings on Freesound, this motivates future work on more advanced semi-supervised methods, such as FixMatch [21] or FlatMatch [22].

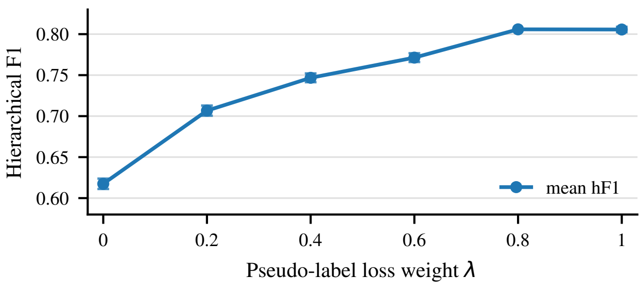

Fig. 1: hF1 score as a function of the pseudo-label loss weight λ in Eq. (9) used in the pretraining stage (Stage 2).

5.3. Impact of the Stage 2 Training-Set Size

To study the effect of the Stage-2 training-set size, we train CLAP models with $\lambda = 1$ on randomly sampled and confidence-filtered subsets of BSD35k-CS. For confidence filtering, examples are selected per class according to their Stage 1 ensemble scores, prioritizing more reliable pseudo-labels while preserving class coverage.

As shown in Fig. 2, performance initially improves for both selection strategies and then largely saturates. Randomly sampled subsets perform marginally better at smaller set sizes and appear to saturate earlier, at around 40% of the data. This may indicate that lower-confidence examples near the teacher's decision boundaries help the student imitate its prediction behavior.

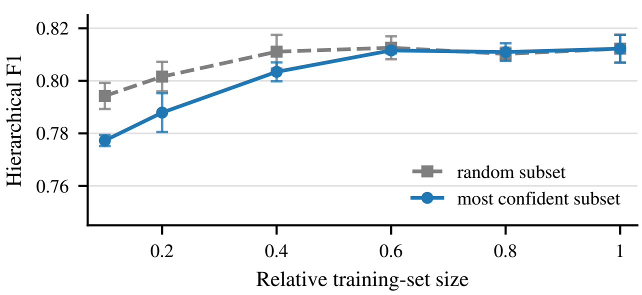

Fig. 2: Stage-2 validation hF1 as a function of the retained fraction of BSD35k-CS. Results compare random subsampling with class-balanced confidence filtering based on Stage 1 ensemble scores.

5.4. Model Error Analysis

We additionally analyze the predictions of the final Stage-3 model on the BSD10k test set. The annotated class appears among the top two predictions for 94.30% of recordings and among the top three for 97.26%, compared with a top-1 accuracy of 83.76%. Even when the top-ranked prediction is incorrect, the classifier often assigns a high score to the annotated class.

The most frequent confusion is between soundscapes from indoor spaces (`ss-i`) and soundscapes from urban outdoor spaces (`ss-u`). The model predicts `ss-i` instead of `ss-u` for 12 examples in the test set and makes the reverse error for 10 examples. These predictions have mean top-1 confidences of 69.0% and 67.1%, respectively, suggesting substantial uncertainty between the two classes. Other recurring errors occur between closely related music classes, including music with multiple instruments (`m-m`) and music with a single instrument (`m-si`), and among several soundscape and sound-effect classes. For example, soundscapes from natural habitats (`ss-n`) are confused with sound effects from natural elements (`fx-n`) in 11 cases, while sound effects from animals (`fx-a`) and `ss-n` are confused in both directions. Overall, the confusion patterns indicate that errors are concentrated among semantically overlapping categories.

We manually inspected all 22 recordings confused between `ss-i` and `ss-u`, classifying each as a model error, likely annotation error, or unclear case. Only 10 appear to be unambiguous model errors. In eight cases, the predicted class was judged to describe the recording more accurately than the reference annotation. Examples include recordings annotated as `ss-u` that appear to have been captured inside a train compartment, museum, store, theater, or processing plant. Conversely, some recordings annotated as `ss-i` contain evidence of an outdoor or urban setting. Four recordings could not be assigned confidently because the location or acoustic context was insufficiently clear.

A recurring source of ambiguity between ss-u and ss-i is the distinction between a recording's physical enclosure and broader semantic environment. Public places such as stores, markets, museums, metro stations, and bazaars are physically indoors, but may contain speech, traffic sounds, or other sounds strongly associated with an urban environment, leading to annotation and model errors. Recordings inside moving vehicles pose a similar problem: the microphone is enclosed, but the soundscape may be dominated by the surrounding urban or transportation environment.

6. CONCLUSION

In this work, we investigated metadata-aware classification and the use of noisy crowdsourced training data for heterogeneous audio classification. Our experiments show that textual metadata provides a strong source of information complementary to the audio signal. In particular, incorporating LLM-derived BST class predictions consistently improves both CLAP- and M2D-based systems, with the best single-model configuration achieving an hF1 of 82.04% on the held-out BSD10k test split. For BSD35k-CS, replacing the original crowdsourced labels with soft predictions from models trained on clean data is substantially more effective than relying directly on the noisy annotations. However, the gains from pseudo-labeled pretraining remain modest, indicating that additional data does not necessarily improve performance when its supervision is derived from the same clean-data models. Finally, the remaining errors are concentrated among semantically related and ambiguously defined BST classes. Future work should investigate stronger semi-supervised learning methods and tools that help crowd annotators select the correct BST category, particularly for classes frequently confused by both models and human annotators.

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